CPTG and the Cosmological Horizon Problem:

Native Origin, Geometric Containment, and Acoustic Projection

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Abstract

The cosmological horizon problem is usually stated as a single-clock causal puzzle: the cosmic microwave background is nearly uniform across the sky even though widely separated regions of the last-scattering surface are not ordinarily assigned enough pre-decoupling light-travel time to exchange heat and equilibrate. Curvature Polarization Transport Gravity (CPTG) has a different cosmological architecture. It contains one native cosmological π-branch and two observational projections, acoustic and luminosity, which share the same geometric phase $N_\pi = \ln a$ but generally accumulate different branch times.

This distinction is central. The acoustic CMB background is not constructed from independently initialized acoustic systems whose temperatures must later be synchronized by photons within one universal clock interval. Instead, the acoustic background is a single-valued projection of the native geometric state. The branch-time distinction is not a claim that photons propagate faster than c; it states that a single observational time coordinate cannot be imposed on all three CPTG response layers.

The paper then performs a Planck-anchored observational-to-native containment test. At the adopted Planck last-scattering phase, the physical radius of the region that becomes the observed last-scattering sphere is $L_\star = 12.72029287\,\mathrm{Mpc}$, while the native curvature-transport radius is $R_C = 23.64711498\,\mathrm{Mpc}$, giving $L_\star/R_C = 0.53792156 < 1$. The acoustic projector, exact within the declared local monomial class, preserves the native background normalization, so the contained region carries one acoustic background datum rather than a collection of independently normalized acoustic patches.

The resulting background-horizon construction therefore rests on three linked statements: one native geometric history, projected branches with distinct time scales, and like-for-like spatial containment at last scattering. The numerical containment value reported here is a Planck-anchored observational-to-native containment test: Planck supplies the adopted last-scattering phase, while the native R_C trajectory and the acoustic comparison-distance construction are CPTG quantities. SH0ES does not enter the primordial containment gate.

1 The Horizon Problem in CPTG

The cosmic microwave background (CMB) is nearly uniform, with residual anisotropies at approximately the 10^{-5} level [1, 2, 3]. In the conventional horizon formulation, widely separated regions of the last-scattering surface are compared within one cosmological time description. If ordinary signals are limited by the light cone and insufficient time is available before decoupling, those regions cannot exchange enough heat to explain why their background temperatures are nearly identical.

The phrase “not enough time” is therefore meaningful only after the relevant clock has been specified. CPTG does not assign one universal observational clock to all of its cosmological response coordinates. Its construction contains one native geometric π-branch and two projections, acoustic and luminosity [5, 6, 7]. They share the same geometric phase $N_\pi = \ln a$, but the projected branches generally accumulate different times along that common phase history.

The CPTG proposal is consequently not a faster-than-light photon or heat-transfer mechanism. The acoustic CMB background is not treated as a collection of independently initialized acoustic cosmologies that must communicate across the sky within one common time interval. It is a projection of one native geometric history. The causal question is therefore reorganized into three concrete checks: whether the native and projected descriptions refer to the same geometric phase, whether the observed last-scattering region is geometrically contained within the relevant native curvature scale, and whether the acoustic background follows uniquely from the native state without an additional primordial acoustic normalization.

These checks have distinct roles. Common phase with different branch clocks addresses the time-coordinate assumption built into the usual causal statement. Geometric containment tests the spatial placement of the future observable last-scattering region. The exact acoustic projector tests whether an independently synchronized acoustic background datum is required. None of these statements is interpreted as superluminal material propagation.

Two levels must also be kept distinct. The background problem concerns the common geometric state on which the CMB monopole and acoustic background are defined. The measured anisotropies are small residual departures from that background and lie outside the background-horizon claim established here.

The derivation below uses CPTG’s own native scale relations. Friedmann equations, a dark-matter component, an inflaton, and a separately imposed cross-branch thermal synchronization law are not used as governing machinery.

The argument proceeds in a defined sequence. Section 2 establishes the common geometric phase, the one-native-branch projection architecture, and the distinct branch clocks. Section 3 identifies the native spatial scale relevant to the horizon question. Section 4 compares that scale directly with the physical size of the future observable last-scattering region. Section 5 then combines containment with the exact acoustic projector. Section 6 develops the accumulated branch-time separation and the Planck/SH0ES comparison coordinates. The final sections keep observational checks and direct failure conditions separate from the background construction.

2 Common Geometric Phase: One Native π-Branch and Two Projections

The first step is to specify what is shared by the branches and what is not. CPTG uses one geometric stage variable for the native history and its observational projections, but it does not force those projections to accumulate the same branch-specific time coordinate. This distinction is essential: common phase identifies corresponding states, whereas the branch clocks describe how each response layer accumulates its integrated time coordinate along that shared sequence. The construction does not assert that physical time itself is different between branches.

Let

$$N_\pi \equiv \ln a \tag{1}$$

denote the shared geometric phase label. For every branch,

$$\frac{dN_\pi}{dt_b} = H_b(N_\pi), \qquad H_b(N_\pi)\, dt_b = dN_\pi, \qquad b \in \{\pi, \mathrm{ac}, L\}. \tag{2}$$

The same value of $a = e^{N_\pi}$ therefore identifies the same geometric stage, not the same clock reading. The branch-time functions are not identical and no equality of their clock readings is imposed:

$$t_b(a) \not\equiv t_{b'}(a), \qquad b \neq b'. \tag{3}$$

Thus the acoustic branch can reach a common geometric phase at a different accumulated time from the native branch. The branch clocks are related through the common phase increment dN_π; they are not forced to be equal. This is the precise sense in which the conventional statement that widely separated regions had "insufficient time" cannot be transferred to CPTG without first specifying which branch clock is being used. The distinction does not modify the local light-speed limit; it distinguishes the native geometric history from the time coordinates assigned to its observational projections.

The three-branch response structure is organized as

$$\text{one native } \pi\text{-branch} \quad \longrightarrow \quad \begin{cases} \text{acoustic projection,} \\ \text{luminosity projection.} \end{cases} \tag{4}$$

The arrows in Eq. (4) are projection maps, not material transport between different universes.

2.1 Native π-branch

Everything downstream begins with the native background. The purpose of this subsection is therefore only to fix the single geometric history that will later be projected into acoustic and luminosity coordinates. No observational branch is introduced at this stage.

The native expansion function is

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \qquad \mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \tag{5}$$

with

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a), \qquad H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{6}$$

The nonconstant native contribution is

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \tag{7}$$

There is one background function $X_\pi(a)$. Introducing an arbitrary patch normalization $C_\pi(\mathbf{x})$ would replace the single native state with a collection of independently initialized native realizations and would not represent the CPTG native-branch construction.

2.2 Exact acoustic projection

Once the native state is fixed, the next question is whether the acoustic sector requires an additional primordial normalization. The answer is determined by the projector itself. Rather than assigning the acoustic branch an independent starting amplitude, CPTG maps the nonconstant native contribution directly into the acoustic response.

The acoustic exponent and projection ratio are

$$p_{\rm ac} = 3 - \frac{\pi}{100} = 2.968584073464102, \qquad \mathcal{G}_T = \frac{p_{\rm ac}}{\pi} = 0.9449296585513721. \tag{8}$$

Define the acoustic projector by

$$\mathcal{P}_{\rm ac}(X) \equiv \mathcal{T}_\pi^{1-\mathcal{G}_T} X^{\mathcal{G}_T}. \tag{9}$$

Within the declared class of local monomial maps $\mathcal{P}(X) = AX^\gamma$, requiring the native term $\mathcal{T}_\pi a^{-\pi}$ to map to exponent $p_{\rm ac}$ while inheriting the coefficient $\mathcal{T}_\pi$ exactly fixes $\gamma = p_{\rm ac}/\pi = \mathcal{G}_T$ and $A = \mathcal{T}_\pi^{1-\mathcal{G}_T}$. Accordingly, "exact acoustic projector" in this paper means exact within that declared local monomial class; it does not assert uniqueness over every broader non-monomial, nonlocal, history-dependent, or derivative-dependent representation.

Applying it to Eq. (7) gives

$$\begin{aligned} X_{\rm ac}(a) &= \mathcal{P}_{\rm ac}[X_\pi(a)] && (10)\\ &= \mathcal{T}_\pi^{1-\mathcal{G}_T}\left(\mathcal{T}_\pi a^{-\pi}\right)^{\mathcal{G}_T} && (11)\\ &= \mathcal{T}_\pi a^{-\pi\mathcal{G}_T} && (12)\\ &= \mathcal{T}_\pi a^{-p_{\rm ac}}. && (13) \end{aligned}$$

Accordingly,

$$E_{\rm ac}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}. \tag{14}$$

The acoustic rate normalization associated with this projection is:

$$H_{0,\rm ac} = H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{15}$$

Equation (13) is the central background result. The acoustic branch changes the response exponent while preserving the native weight $\mathcal{T}_\pi$ exactly. It therefore has no independent primordial background normalization that must be synchronized with the native branch or with other spatial patches.

This completes the primordial branch-mapping part of the argument. The luminosity branch belongs to the same common-phase hierarchy but is a late observational response and does not participate in establishing primordial acoustic coherence.

2.3 Luminosity projection in the common-phase hierarchy

The luminosity branch carries the native phase history into the local luminosity/time response. Its detailed clock map and SH0ES-conditioned comparison normalization are treated in Section 6, after the native containment and acoustic-projection arguments have been combined.

3 Native Geometric Scale: The Curvature-Transport Radius

The curvature-transport radius is the native length scale used for the containment test. It should not be confused with an acoustic propagation distance or with a conventional curvature radius imported from another cosmological framework. Here it is evaluated directly on the same native π-branch defined above, so the phase, expansion, native acceleration, and curvature-transport length coordinates all refer to one underlying geometric state.

On the native branch, the epoch-dependent native cosmological acceleration coordinate is

$$a_\pi(a) = a_{\pi,0}E_\pi(a), \qquad a_{\pi,0} = 5.568327996831708\times10^{-10}\,\mathrm{m\,s^{-2}}. \tag{16}$$

Here $a_\pi(a)$ is the universe-scale native acceleration coordinate. It is distinct from the object-dependent structural acceleration scale $a_\star$; $a_\star$ does not enter the cosmological horizon construction.

The CPTG curvature-transport radius follows from the native cosmological curvature-transport relation

$$R_C(a) = \frac{48\pi c^2}{a_\pi(a)} = \frac{R_{C,0}}{E_\pi(a)}, \qquad R_{C,0} = 788.7823091477\,\mathrm{Gpc}. \tag{17}$$

Here R_C is the native CPTG curvature-transport radius. It is not an ordinary FLRW curvature radius, a photon mean free path, or a dark-matter scale.

For the horizon calculation, R_C is the characteristic radial extent of one connected native curvature-transport domain: a region whose background state is described by the same native solution at a fixed geometric phase. Both $R_C(N_\pi)$ and the future-observable radius $L_\star(N_\pi)$ are radial measures referred to the same native cosmological reference origin/worldline in the CPTG cosmological construction [5, 6]. The containment comparison is therefore concentric within that construction; it is not a comparison of arbitrarily displaced spheres. The calculation determines whether the entire future-observable last-scattering radius lies within that native curvature-transport domain. It does not interpret R_C as a photon travel distance or require a signal to cross the domain after emergence.

The native acceleration and expansion normalizations satisfy

$$a_{\pi,0} = cH_0^{(\pi)}\sqrt{\mathcal{B}_\pi}, \tag{18}$$

so the branch obeys the invariant

$$H_\pi(a)R_C(a) = H_0^{(\pi)}R_{C,0} = \frac{48\pi c}{\sqrt{\mathcal{B}_\pi}}. \tag{19}$$

The expansion rate and curvature radius are therefore reciprocal realizations of the same native geometric state.

Temperature is used only as an ordinary epoch label,

$$\frac{T}{T_0} = a^{-1} = 1 + z, \qquad T_0 = 2.72548\,\mathrm{K}. \tag{20}$$

It does not define another CPTG branch or a common synchronization clock.

The native radius is thus specified as a function of phase without introducing a second causal scale or a separate acoustic horizon.

4 Planck-Anchored Observational-to-Native Last-Scattering Containment

Containment is a like-for-like geometric comparison. The observable quantity is the physical radius of the future last-scattering sphere at a given phase; the native quantity is the curvature-transport radius at that same phase. Comparing radius with radius avoids turning an endpoint-to-endpoint traversal distance into the criterion for whether the entire observed region belongs to one native domain.

For the Planck-anchored observational-to-native containment test, use the Planck last-scattering redshift employed by the CPTG acoustic comparison construction,

$$z_\star = 1089.92, \qquad a_{\rm dec} = \frac{1}{1+z_\star} = 9.1665750009 \times 10^{-4}, \tag{21}$$

using the Planck last-scattering value [3, 6]. This $z_\star$ is an observational comparison anchor, not a CPTG-native derivation of recombination or a standard-cosmology time input. In the CPTG acoustic comparison construction, the corresponding comoving distance is

$$\chi_\star \equiv \chi_{\rm ac}(z_\star) = \frac{c}{H_0^{(\pi)}} \int_0^{z_\star} \frac{dz}{E_{\rm ac}(z)} = 13876.8219\,\text{Mpc}. \tag{22}$$

The dimensional prefactor is the native normalization $H_0^{(\pi)}$, while the acoustic projection enters through $E_{\rm ac}(z)$ in the integrand [6]. The resulting $\chi_\star$ is therefore a CPTG acoustic comparison-distance coordinate evaluated to the adopted Planck phase; the calculation does not import a standard-ΛCDM expansion-time history as governing machinery. For the fixed future-observable comoving radius $\chi_\star$, define its physical radius at any common phase by

$$L_\star(a) \equiv a\,\chi_\star. \tag{23}$$

Although $\chi_\star$ is obtained through the acoustic comparison layer, its conversion to physical radius is evaluated at the common geometric phase: $L_\star(N_\pi) = e^{N_\pi}\chi_\star$. The containment calculation therefore compares $L_\star(N_\pi)$ and $R_C(N_\pi)$ at corresponding geometric states. It never sets $t_{\rm ac} = t_\pi$. At decoupling this becomes

$$L_\star(z_\star) = a_{\rm dec}\chi_\star = 12.72029287\,\text{Mpc}. \tag{24}$$

The native branch at the same phase gives

$$E_\pi(a_{\rm dec}) = 3.3356386600 \times 10^4, \tag{25}$$

and therefore

$$R_C(z_\star) = \frac{R_{C,0}}{E_\pi(a_{\rm dec})} = 23.64711498\,\text{Mpc}. \tag{26}$$

The Planck-anchored observational-to-native containment ratio is

$$\Lambda_R(z_\star) \equiv \frac{L_\star(z_\star)}{R_C(z_\star)} = 0.53792156 < 1. \tag{27}$$

Equivalently,

$$\frac{2L_\star}{2R_C} = 0.53792156. \tag{28}$$

Comparing the full diameter $2L_\star$ with the single radius R_C mixes unlike geometric quantities and is not used as the closure criterion.

Reversing the same native trajectory, the equality

$$L_\star(z_{\rm enter}) = R_C(z_{\rm enter}) \tag{29}$$

occurs at

$$z_{\rm enter} = 3231.635355, \qquad T_{\rm enter} = 8810.483007\,\text{K}, \tag{30}$$

with

$$R_C(z_{\mathrm{enter}}) = L_\star(z_{\mathrm{enter}}) = 4.292727257\,\mathrm{Mpc}. \tag{31}$$

This equality marks when the future observable radius becomes smaller than one native R_C. It is not the beginning of coherence. The native background already consists of one geometric state, and the acoustic state is already its projection.

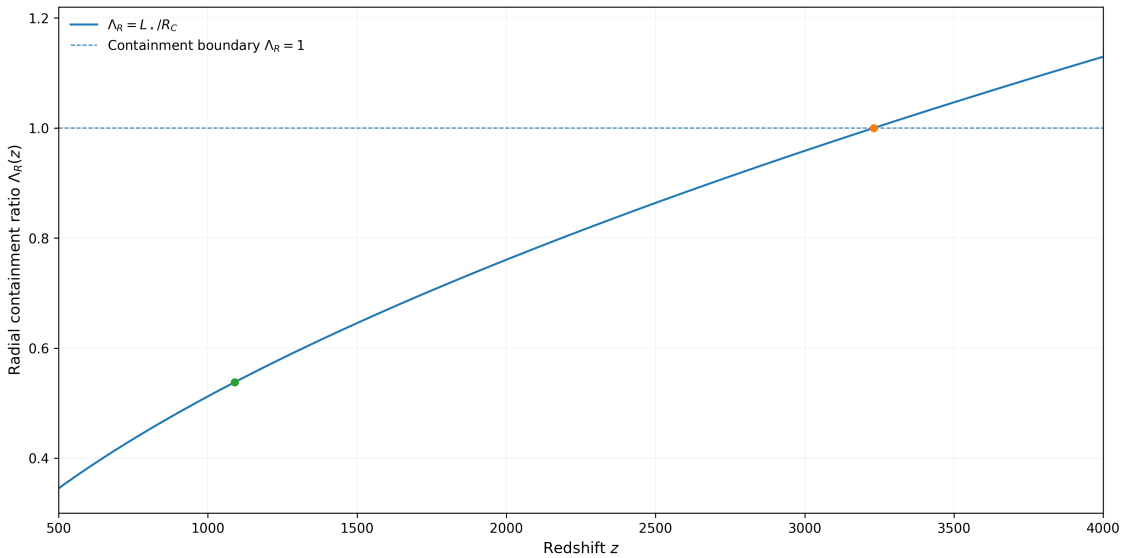

Figure 1: Native radial containment ratio $\Lambda_R(z) = L_\star(z)/R_C(z)$ over the interval shown. The horizontal dashed line is the like-for-like containment boundary $\Lambda_R = 1$. The trajectory crosses that boundary at $z_{\mathrm{enter}} = 3231.635355$, where $L_\star = R_C = 4.292727257\,\mathrm{Mpc}$. At the adopted Planck last-scattering phase $z_\star = 1089.92$, the ratio has fallen to $\Lambda_R = 0.53792156$, corresponding to $L_\star = 12.72029287\,\mathrm{Mpc}$ inside $R_C = 23.64711498\,\mathrm{Mpc}$.

The containment result by itself is only spatial. Background closure requires combining it with the projection result from Section 2.

5 From Containment to Background Horizon Closure

The containment and projection calculations answer different questions. Containment establishes that the relevant last-scattering region fits inside one native curvature domain; projection establishes how an observational branch is assigned to that native state. Neither statement should be asked to do the work of the other. Their combination is what changes the horizon problem from a post-emergence synchronization problem into a common-origin problem.

The native branch supplies one background state on the connected native domain. CPTG cosmology specifies one connected native solution followed through the phase variable N_π, rather than an ensemble of independently normalized patch solutions [5, 6]. The acoustic projector does not create native homogeneity; it maps the specified native background into the acoustic response. Denote the native background by $X_\pi^{(0)}(a)$.

Within that single-native-solution background premise, the native field has one background value at each phase,

$$\nabla_i X_\pi^{(0)} = 0. \tag{32}$$

For the explicit local acoustic projector in Eq. (9),

$$X_{\rm ac}^{(0)}(a) = \mathcal{P}_{\rm ac}[X_\pi^{(0)}(a)], \tag{33}$$

so the ordinary chain rule gives

$$\nabla_i X_{\rm ac}^{(0)} = \mathcal{P}'_{\rm ac}[X_\pi^{(0)}]\nabla_i X_\pi^{(0)} = 0. \tag{34}$$

This result is specific to the local scalar projector actually derived here; no broader claim is made for arbitrary nonlocal, history-dependent, or derivative-dependent projection operators. Equation (13) also proves that the acoustic normalization is inherited from the native branch.

This step does not derive native homogeneity by local causal equilibration. The single-native-background premise is part of the CPTG cosmological architecture; the calculation tests whether that already-defined native background, together with the Planck-anchored observational-to-native containment test and the acoustic projector, removes the need for independently initialized acoustic patches.

Equations (27) and (34) answer different parts of the problem:

1. $L_\star/R_C < 1$ establishes that the observed last-scattering radius lies inside one native CPTG curvature domain at decoupling.
2. The one-native-state architecture establishes that this domain is not assembled from independently normalized background patches.
3. The exact acoustic projector establishes that the acoustic background is a single-valued response of the native state.

No signal must traverse the last-scattering diameter to make separately initialized acoustic backgrounds agree, because there are no independently initialized acoustic backgrounds in this construction. The coherence is inherited algebraically from the native state rather than produced by post-emergence material equilibration.

The remaining difference between the branches is not their origin but their subsequent response.

6 Branch Clocks and Forward Projection Separation

Branch-specific time enters only after the common-origin construction is established. It is therefore not a mechanism used to solve the primordial horizon problem. Its purpose is to describe how the acoustic and luminosity responses accumulate time differently as the shared native phase advances. In this sense, clock separation is a prediction of projection, not evidence that the branches have become disconnected.

Equation (2) gives the essential rule: equal N_π means equal geometric phase, not equal branch time. Using the common boundary convention $t_b(a \to 0) = 0$, the accumulated time on each branch is

$$t_b(a) = \int_0^a \frac{da'}{a' H_b(a')}, \qquad b \in \{\pi, \mathrm{ac}, L\}. \tag{35}$$

The accumulated separation from native time is

$$\Delta t_{b\pi}(a) = t_b(a) - t_\pi(a) = \int_0^a \left[\frac{1}{H_b(a')} - \frac{1}{H_\pi(a')} \right] \frac{da'}{a'}, \tag{36}$$

and hence

$$\frac{d\Delta t_{b\pi}}{dN_\pi} = \frac{1}{H_b} - \frac{1}{H_\pi}. \tag{37}$$

The sign of Eq. (37) determines whether the accumulated separation grows or contracts: $H_b < H_\pi$ gives $d\Delta t_{b\pi}/dN_\pi > 0$, $H_b > H_\pi$ gives $d\Delta t_{b\pi}/dN_\pi < 0$, and a rate crossing $H_b = H_\pi$ gives a stationary separation. Defining the branch maps

$$\mathcal{A}(N_\pi) \equiv \frac{H_{\mathrm{ac}}(N_\pi)}{H_\pi(N_\pi)}, \qquad \mathcal{L}(N_\pi) \equiv \frac{H_L(N_\pi)}{H_\pi(N_\pi)}, \tag{38}$$

gives the exact clock relations

$$\frac{dt_{\mathrm{ac}}}{dt_\pi} = \frac{1}{\mathcal{A}(N_\pi)}, \qquad \frac{dt_L}{dt_\pi} = \frac{1}{\mathcal{L}(N_\pi)}. \tag{39}$$

The equal phase increment dN_π therefore identifies corresponding states without making the branch clocks equal. The different clock readings at the same geometric stages are especially transparent at the entry and decoupling phases. The acoustic times in Table 1 are calculated from the exact acoustic projection E_{ac} in Eq. (14), not from the separate acoustic-density comparison background defined below.

Common geometric phase	Native time t_π	Exact acoustic time t_{ac}	Role
$z_{\mathrm{enter}} = 3231.635355$	48,802 yr	106,889 yr	Radial-entry phase, where $L_\star = R_C$.
$z_\star = 1089.92$	268,835 yr	536,006 yr	Planck last-scattering phase.

Table 1: Native and exact-acoustic clock readings at two shared geometric phases. Each row represents one value of N_π and therefore one geometric stage; the branch clocks assign different accumulated times to that stage.

The table makes the common-phase construction explicit: both branches are evaluated at the same scale-factor value $a = e^{N_\pi}$ even though the clock readings differ. Equivalently, at fixed common phase,

$$t_\pi(a) \neq t_{\mathrm{ac}}(a), \qquad N_\pi = \ln a. \tag{40}$$

6.1 Luminosity/time projection and calibrated normalization

The luminosity branch now enters as the late-time counterpart of the same common-phase construction. At fixed N_π, write

$$H_L(N_\pi) = \mathcal{L}(N_\pi)H_\pi(N_\pi), \qquad \Xi_L(N_\pi) \equiv \frac{1}{\mathcal{L}(N_\pi)}. \tag{41}$$

The projected luminosity time therefore satisfies

$$\frac{dt_L}{dt_\pi} = \Xi_L(N_\pi) = \frac{1}{\mathcal{L}(N_\pi)}, \qquad H_L\, dt_L = H_\pi\, dt_\pi = dN_\pi. \tag{42}$$

Luminosity time is a projection of native time, not an additional independent cosmological clock.

To avoid confusing the present phase $N_\pi = 0$ with the singular limit $a = 0$, define

$$\mathcal{L}_0 \equiv \mathcal{L}(N_\pi = 0) = \mathcal{L}(a = 1), \tag{43}$$

$$\Xi_{L,0} \equiv \Xi_L(N_\pi = 0) = \Xi_L(a = 1) = \mathcal{L}_0^{-1}. \tag{44}$$

Using the four-anchor local distance-ladder value [4],

$$H_{0,L} = 73.17 \pm 0.86\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{45}$$

$$\mathcal{L}_0 = \frac{H_{0,L}}{H_0^{(\pi)}} = 1.054075942846 \pm 0.012389029805, \tag{46}$$

$$\Xi_{L,0} = 0.948698247775 \pm 0.011150478244. \tag{47}$$

These are observationally conditioned luminosity/time coordinates, not independent native predictions. The exact acoustic projector above is the branch relation used in the horizon argument. For the late-time branch-rate visualization, CPTG also uses a separate acoustic-density comparison background. It is introduced here only to display the forward separation of observational response coordinates and should not be confused with the exact primordial acoustic projector. The comparison weights are

$$\Omega_{m,\mathrm{ac}} = \frac{p_{\mathrm{ac}}}{3\pi} = 0.314976552850, \qquad \Omega_{\Lambda,\mathrm{ac}} = 1 - \Omega_{m,\mathrm{ac}} = 0.685023447150, \tag{48}$$

The symbols $\Omega_{m,\mathrm{ac}}$ and $\Omega_{\Lambda,\mathrm{ac}}$ are CPTG acoustic comparison weights. They do not introduce matter, dark-matter, or dark-energy components into the native CPTG dynamics. With these comparison weights,

$$E_{\mathrm{ac},\rho}(a)^2 = \Omega_{\Lambda,\mathrm{ac}} + \Omega_{m,\mathrm{ac}}a^{-p_{\mathrm{ac}}}. \tag{49}$$

This is a CMB/acoustic density-weighted comparison background and is distinct from the exact acoustic projector in Eq. (14). The SH0ES-conditioned luminosity/time comparison realization used in the late-time trajectory is

$$H_L^{\mathrm{anchor}}(a) = H_{0,L}E_{\mathrm{ac},\rho}(a). \tag{50}$$

Because luminosity time is a projection of native time, this realization has the explicit clock map

$$\Xi_L^{\text{anchor}}(a) = \frac{H_\pi(a)}{H_L^{\text{anchor}}(a)} = \frac{H_0^{(\pi)}}{H_{0,L}} \left[\frac{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}{\Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}} \right]^{1/2}, \tag{51}$$

which recovers $\Xi_{L,0} = 0.948698247775 \pm 0.011150478244$ at $a = 1$ ($N_\pi = 0$). Thus the full-history luminosity/time comparison is an explicit SH0ES-conditioned projection of the native phase history and is used here only to display the late branch-time response.

Figures 2 and 3 show two complementary parameterizations of the same branch architecture. Figure 2 plots accumulated branch time against the shared geometric phase, so vertical alignment means the same N_π. Figure 3 instead plots each branch rate against that branch's own integrated time, so equal horizontal position there does not mean equal geometric phase.

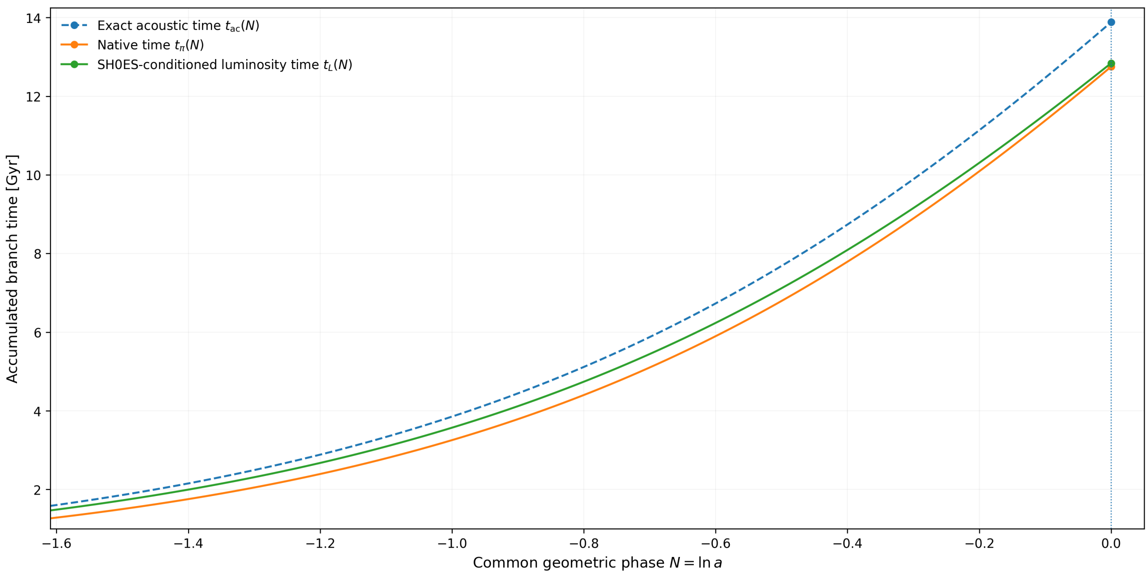

Figure 2: Common-phase branch-time trajectories over $0.20 \le a \le 1$. Every trajectory is evaluated against the same horizontal coordinate $N_\pi = \ln a$. The exact acoustic trajectory uses E_{ac} from Eq. (14); the native trajectory uses E_π; and the luminosity trajectory uses the SH0ES-conditioned comparison realization of Eq. (51). The vertical dotted guide at $N_\pi = 0$ marks the present common phase. At that phase the integrated times are approximately $t_\pi = 12.7541\,\text{Gyr}$, $t_L = 12.8431\,\text{Gyr}$, and $t_{\text{ac}} = 13.8852\,\text{Gyr}$.

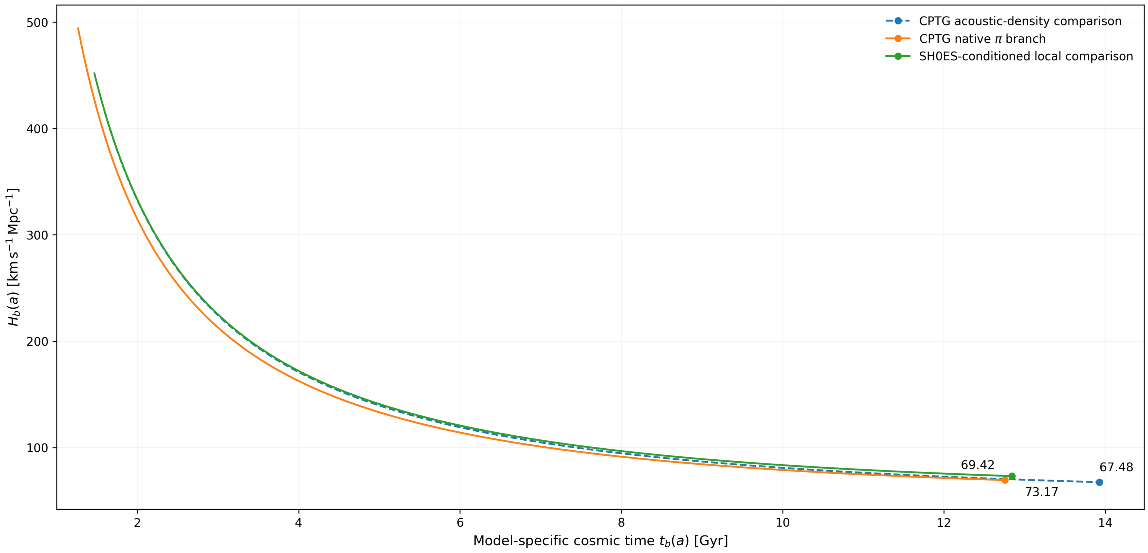

Figure 3: Late-time branch-rate trajectories over $0.20 \le a \le 1$. The native π branch, acoustic-density comparison, and SH0ES-conditioned luminosity/time comparison are evaluated on the same scale-factor sequence but each is plotted against its own integrated branch time $t_b(a)$. Horizontal alignment in this figure therefore does not denote equal geometric phase. The acoustic-density curve uses $E_{\mathrm{ac},\rho}$ rather than the exact primordial acoustic projector E_{ac}, while the luminosity curve is the endpoint-conditioned realization in Eq. (51). The endpoint labels 67.48, 69.42, and 73.17 are the corresponding present rates in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The late branch hierarchy is

$$H_{0,\mathrm{ac}} < H_0^{(\pi)} < H_{0,L}, \tag{52}$$

or numerically

$$67.4777967351 < 69.4162507897 < 73.17 \qquad (\mathrm{km\,s^{-1}\,Mpc^{-1}}). \tag{53}$$

The acoustic-to-luminosity endpoint separation is

$$\frac{H_{0,L}}{H_{0,\mathrm{ac}}} - 1 = 0.0843566853 = 8.43566853\%. \tag{54}$$

The same structure therefore supports an early common native origin together with branch-dependent late observational response rates; the accumulated clock separations need not be monotonic.

The horizon closure itself has not depended on fitting those late endpoints. They are useful instead as consistency checks on whether the same projection architecture remains viable when translated into actual observational coordinates.

7 Planck and SH0ES Comparison Checks

The observational comparisons in this section are restricted to Planck and SH0ES. Planck supplies the adopted last-scattering phase and the CMB comparison quantities used by the acoustic layer, while SH0ES supplies the local distance-ladder endpoint used for the luminosity branch. The native $a_\pi(a)$ and $R_C(a)$ trajectories are not defined by either dataset; however, the reported numerical last-scattering containment value is explicitly a Planck-anchored observational-to-native containment test because it is evaluated at the Planck-supplied $z_\star$. Data-conditioned quantities are identified as such rather than presented as independent native predictions.

7.1 Acoustic angular-scale consistency check

The Planck acoustic-angle value used for comparison is $100\theta_\star^{\rm obs} = 1.04110$ [3]. With the CPTG acoustic distance

$$\chi_\star = 13876.8219\,\mathrm{Mpc}, \tag{55}$$

the corresponding implied acoustic ruler is

$$r_{\rm ac} = 144.4716\,\mathrm{Mpc}. \tag{56}$$

Reinserting this ruler and distance gives

$$100\theta_\star^{\rm recon} = 100\frac{r_{\rm ac}}{\chi_\star} = 1.04110005188. \tag{57}$$

This is an arithmetic reconstruction check of the adopted angular anchor, not an independent prediction of $\theta_\star$. The model-inferred ruler consistency comparison is

$$\frac{r_{\rm ac}^{\rm CPTG}}{r_\star^{\rm ref}} = \frac{144.4716}{144.43} = 1.000288, \tag{58}$$

where 144.43 Mpc is retained only as the Planck-inferred comparison ruler used in the CPTG comparison-coordinate construction [6, 3].

7.2 Present branch endpoints

Branch	H_0 [km s^{-1} Mpc^{-1}]	Classification	Comparison role
Acoustic	67.4777967351	Derived comparison coordinate	Planck-facing acoustic/CMB endpoint generated by the acoustic projection defined above.
Native π	69.4162507897	Derived branch quantity	Native CPTG present-rate normalization.
Luminosity	73.17 ± 0.86	Data-conditioned comparison coordinate	SH0ES-conditioned local endpoint; $\mathcal{L}_0 = 1.054075942846 \pm 0.012389029805$.

Table 2: Present endpoint values of the native and projected CPTG response coordinates. The classification column distinguishes native, projected, and observationally conditioned quantities; the rows are not three independent cosmological fits.

The acoustic endpoint is directly comparable to the Planck 2018 base-ΛCDM CMB inference,

$$H_{0,\rm ac}^{\rm CPTG} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad H_0^{\rm Planck} = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{59}$$

while the luminosity endpoint is conditioned specifically on the SH0ES local distance ladder [3, 4]. CPTG interprets this Planck–SH0ES difference as branch response separation from one native evolution. It is not attributed to an added dark-matter component, a dark-energy fluid, or an alteration of the native branch normalization.

The SH0ES endpoint and late branch-rate comparisons are not used to define the primordial horizon closure. Planck has the narrower but explicit role of supplying the observational last-scattering anchor for the Planck-anchored observational-to-native containment test. Residual CMB anisotropies lie outside the background-horizon claim of this paper.

8 Falsifiability and Scope

The horizon claim is deliberately narrow, so its direct failure conditions are equally narrow. The construction fails if the native branch architecture fails, if the like-for-like containment condition fails, or if the exact acoustic projection fails.

8.1 Native-geometry failure

The native geometric layer fails if the branch does not satisfy its reciprocal curvature relation,

$$R_C(a) = \frac{R_{C,0}}{E_\pi(a)}, \tag{60}$$

or if the single connected native background is not a valid solution of the CPTG cosmological construction.

8.2 Horizon-containment failure

The Planck-anchored observational-to-native containment test fails if the future-observable last-scattering region is not contained within the native curvature-transport domain at the adopted Planck phase. In the present calculation the direct gate is

$$\frac{L_\star(z_\star)}{R_C(z_\star)} < 1. \tag{61}$$

8.3 Acoustic-projection failure

The projection layer fails if the acoustic response does not satisfy

$$X_{\rm ac} = \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi^{\mathcal{G}_T}, \tag{62}$$

with inherited native normalization, or if the Planck-facing acoustic comparison coordinates are inconsistent with the comparison construction defined here. Equation (57) remains an arithmetic reconstruction of the adopted Planck angular anchor and is not counted as an independent prediction.

The SH0ES endpoint has a different role: it calibrates the late luminosity/time comparison coordinate and is not part of the primordial horizon-closure gate. The observational comparison scope of this paper is limited to Planck and SH0ES. The native branch and acoustic projection supply the CPTG theory structure, while the numerical containment result is specifically the Planck-anchored observational-to-native containment test defined above.

9 Conclusion

The horizon problem is often presented as a single-clock demand for communication between distant primordial regions. CPTG changes that organization at the outset: one native cosmological π-branch supplies the geometric history, while the acoustic and luminosity branches are projections that share its phase sequence but generally accumulate different times. The construction therefore does not begin with independently initialized acoustic regions that must exchange heat within one universal clock interval.

The exact acoustic projector

$$X_{\rm ac} = \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi^{\mathcal{G}_T} = \mathcal{T}_\pi a^{-p_{\rm ac}} \tag{63}$$

preserves the native normalization, so the acoustic background has no independent primordial datum requiring later synchronization.

At the adopted Planck last-scattering phase,

$$R_C = 23.64711498\,\mathrm{Mpc}, \qquad L_\star = 12.72029287\,\mathrm{Mpc}, \qquad \frac{L_\star}{R_C} = 0.53792156 < 1. \tag{64}$$

Within the Planck-anchored observational-to-native containment test, the future-observable last-scattering radius therefore lies inside one native curvature-transport domain. Because its acoustic state is the deterministic projection of the one native background, the near-uniform CMB background does not require cross-R_C thermal equilibration or communication between independent acoustic patches.

The same projection hierarchy naturally permits distinct late-time branch-rate endpoints without requiring monotonic separation of the accumulated branch clocks. The acoustic, native, and luminosity endpoints occupy 67.4778, 69.4163, and $73.17\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, respectively. Equal N_π labels equal geometric phase, while the corresponding branch times are generally unequal; the acoustic branch therefore reaches a common phase on its own projected time coordinate rather than on the native clock. Early common origin and branch-dependent late observational response are two consequences of the same CPTG branch structure.

Within the stated CPTG native-branch architecture and the declared Planck-anchored observational-to-native containment test, the background-homogeneity component of the cosmological horizon problem is closed at the level tested here. In this construction, the conventional synchronization problem is reclassified: one native geometric history generates projected branches with distinct time coordinates, the Planck-anchored future-observable last-scattering region is contained within the native curvature-transport scale at the adopted phase, and the acoustic background is a deterministic projection rather than an independently initialized system requiring later thermal synchronization.

A Numerical Summary

Table 3: Native, projected, observationally conditioned, and derived quantities used in the CPTG background-horizon calculation. The classification column distinguishes theory quantities, observational inputs, comparison coordinates, and arithmetic checks.

Quantity	Defining relation or value	Classification	Role
$\mathcal{B}_\pi$	$1 - 1/\pi$	Native invariant	Native late/geometric weight.
$\mathcal{T}_\pi$	$1/\pi$	Native invariant	Native transport weight.
p_{ac}	$3 - \pi/100 = 2.968584073464102$	Derived branch quantity	Acoustic-projection exponent.
$\mathcal{G}_T$	$p_{\mathrm{ac}}/\pi = 0.9449296585513721$	Derived branch quantity	Exact acoustic projection ratio.
$H_{0,\mathrm{ac}}$	$H_0^{(\pi)}\sqrt{\mathcal{G}_T}$ $= 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived comparison coordinate	Planck-facing acoustic/CMB endpoint.
$H_0^{(\pi)}$	$69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Derived branch quantity	Native present-rate normalization.
$H_{0,L}$	$73.17 \pm 0.86\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Data-conditioned comparison coordinate	SH0ES-conditioned luminosity endpoint.
$\mathcal{L}_0$	$1.054075942846 \pm 0.012389029805$	Data-conditioned comparison coordinate	Present luminosity-rate factor relative to native.
$\Xi_{L,0}$	$0.948698247775 \pm 0.011150478244$	Data-conditioned comparison coordinate	Reciprocal present luminosity-rate factor.
$R_{C,0}$	$788.7823091477\,\mathrm{Gpc}$	Derived branch quantity	Present native curvature-transport scale.
$z_\star$	1089.92	Observational input	Planck anchor for the observational-to-native containment test.
$E_\pi(a_{\mathrm{dec}})$	3.3356386600×10^4	Derived branch quantity	Native expansion value at the adopted decoupling phase.
$R_C(z_\star)$	$23.64711498\,\mathrm{Mpc}$	Derived branch quantity	Native curvature-transport radius at decoupling.
$L_\star(z_\star)$	$12.72029287\,\mathrm{Mpc}$	Derived comparison coordinate	Physical future-observable radius at the same phase.
$L_\star/R_C$	0.53792156	Derived comparison coordinate	Planck-anchored observational-to-native containment result.
z_{enter}	3231.635355	Derived comparison coordinate	Phase at which $L_\star = R_C$.
$100\theta_\star^{\mathrm{recon}}$	1.04110005188	Arithmetic check	Arithmetic reconstruction of the adopted Planck angular anchor.

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